# Project Development and Completion Tracker

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Porject ID: | LYCORE107 | Class | LY - CORE 1 | Percentage of Project Completion: | 100% | Copyright Status: | NA | Technnology Transfer: | NO |
| Project Domain (Don't Add New) | ntelligence (AI) & Machine Lear | Source of Problem Statement | Hackathon ... | Map Sustainable Development Goal: (Don't add New) | 14 Life Below Water | Paper Publication Status: | Prepared | Acheivements: | _Type Your Answer_ 1. 2. |
| Enter Problem Statement | Detecting oil spills at marine environment using Automatic Identification System (AIS) and satellite datasets | | | | | GitHub Link: | https://github.com/DheerajNair123/Oi | Meeting Count | 11 |

| | Enrollment Number | Name of Studnets | Class | Contact Number | Email ID | Tech Skill 1 | Tech Skill 2 | Tech Skill 3 | Global Certification 1 | Global Certification 2 |
|---|---|---|---|---|---|---|---|---|---|---|
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| | | | | | | | | | | |
| Project Guide | Prof. Vilas Khedekar | | | | | | | | | |
| Industry Mentor | | | | | | | | | | |

| | Review Date | Comment by Jury/ Mentor | Reivew Link | | Technology Stack | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|
| Reivew-1 | 19-08-25 | Changes in topic | | | Front-end (Client-side) | | | | | |
| Reivew-2 | | | | | Back-end (Server-side): | | | | | |
| Review-3 | | | | | Database Layer: | | | | | |
| | | | | | Development Tools: | | | | | |
| | | | | | Deployment and Infrastructure: | | | | | |
| | | | | | Specialization Inlinement: | | | | | |

**Epic:** As a research scientist, I want to retrieve raw AIS and satellite data for specific regions and timeframes, so I can conduct further analysis and model development.

Story 1: Receive real-time alerts when an oil spill is detected, in order to enable a rapid response and minimize environmental impact.
Story 2: Access historical oil spill data, to analyze trends and support the development of preventative strategies.
Story 3: Receive notifications when a vessel is near a detected oil spill, to allow for route adjustments and avoid potential contamination.

Task1: Collect & preprocess AIS and satellite datasets.
Task2: Train & fine-tune model.
Task3: Build cloud-based platform.
Task4: Create user-friendly interface.
Task5: Test with diverse historical and simulated oil spill datasets.
Task6: Reduce false positives/negatives in spill detection.
Task7: Enable easy integration.

**Acceptance Criteria**

1. ≥90% detection accuracy.
2. Supports real-time processing.
3. Simple upload/streaming.
4. Clear, user-friendly results.

**This is actual Devlopment planning and tracking. Don't add documentation, survey, topic finalization, etc. activities.**

| Sprint No. | Task Name | Sub Tasks | Task Status | Assigned To | Assigned Date | Deadline | Start Date | Completion Date | Completion Status |
|---|---|---|---|---|---|---|---|---|---|
| **Sprint 1** | Collect & preprocess AIS and satellite datasets. | Collect AIS dataset | Complete | Dheeraj | 21-08-25 | **29-08-25** | **21-08-25** | **28-08-25** | 100% |
| 18th Aug 2025 | | Collect Satellite Dataset | Complete | Dheeraj, Kunal | 21-08-26 | **29-08-26** | **21-08-26** | **28-08-26** | 100% |
| 29th Aug 2025 | | Preprocess AIS dataset | Complete | Dheeraj, Rugved | 21-08-27 | **29-08-27** | **21-08-27** | **28-08-27** | 100% |
| | | Preprocess Satellite dataset | Complete | Kunal | 21-08-28 | **29-08-28** | **21-08-28** | **28-08-28** | 100% |
| **Sprint 2** | Train & fine-tune model. | Build An Algorithm | Complete | Kunal, Dheeraj | **01-09-25** | **14-09-25** | **02-09-25** | **16-09-25** | 100% |
| 1st Sep 2025 | | Finalize an Algorithm | Complete | Kunal, Rugved | **01-09-25** | **14-09-25** | **10-09-25** | **16-09-25** | 100% |
| 14th Sep 2025 | Model Validation | Test accuracy of the model | Complete | Kunal | **01-09-25** | **14-09-25** | **2025-11-09** | **2025-11-09** | 100% |
| | | Using Different Images | Complete | Rugved | **02-09-25** | **14-09-25** | **2025-12-09** | **2025-12-09** | 100% |
| **Sprint 3** | Build an GUI | Search and finalize one alert typ | Complete | Rugved, Kunal | **22-09-25** | **2025-05-10** | **24-09-25** | **2025-01-09** | 100% |
| 22nd Sep 2025 | | Implement alert mechanism | Complete | Rugved, Dheeraj | **25-09-25** | **2025-05-10** | **27-09-25** | **2025-01-10** | 100% |
| 5th Oct 2025 | | Test with Different Datasets | Complete | Dheeraj | **25-09-25** | **2025-05-10** | **2025-01-10** | **2025-05-10** | 100% |
| | | Get Accuracy | Complete | Rugved | **2025-01-10** | **2025-05-10** | **2025-04-10** | **2025-05-10** | 100% |
| **Sprint 4** | Integrating Frontend | Making Of Frontend | Complete | Rugved | **2025-06-10** | **2025-07-10** | **2025-01-10** | **2025-01-10** | 100% |
| 6th Oct 2025 | | | Complete | | | | | | 100% |
| 17th Oct 2025 | | Integrating with the model | Complete | Kunal | **2025-06-10** | **2025-07-10** | **2025-01-10** | **2025-01-10** | 100% |
| | | | Complete | | | | | | 100% |

**Publication Details**

| Sr. No | Paper Title | Name of Journal | Year | Authors | URL | DOI | Volume | Page no. | Publisher |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 2 | | | | | | | | | |

**Patent Details**

| Sr. No. | Title | Inventors | Application No. | Patent Number | Filing Country | Subject Category | Filing Date | Publication Date | Publication Status |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |

**Copyright Details**

| Sr. No. | Title of work | Name of Applicants | Registration No. | Dairy Number | Date | Status | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |

**Event and Participations Details**

| Sr. No. | Name of Event | Type of Event | Date | Type of Participation | Details of Prize won | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 2 | | | | | | | | | |

**Add Weekly Meetings Details [ Minimum 12 meetings ]**

| Meeting No. | Date | Attendees with commas | nda points with comi | | Action Items | Assigned to | itus (click if completed) | | |
|---|---|---|---|---|---|---|---|---|---|
| **1** | 19-08-2025 | Kunal , Dheeraj , Rugved,Vilas Si | IdeaSpark | | Review of Ideaspark poster and Track | Kunal | ☑ | | |
| **2** | 21-08-25 | Rugved,Vilas Sir | Topic Finalization | | Changing Topic according to SIH prob | Rugved | ☑ | | |
| **3** | 28-08-25 | Kunal , Dheeraj,Vilas Sir | Updates | | In the original user stories, the staten | Dheeraj | ☑ | | |
| **4** | 04-09-25 | Rugved, Kunal | Updates | | Revise the algorithm | Kunal | ☑ | | |
| **5** | 07-09-25 | Dheeraj, Kunal | Updates | | | Rugved | ☑ | | |
| **6** | 30-09-25 | Kunal , Dheeraj , Rugved,Vilas Si | Progress Update | | Showing Convolution model | Dheeraj | ☑ | | |
| **7** | 1/10/2025 | Kunal , Dheeraj,Vilas Sir | CNN Model | | Build GUI | Kunal | ☑ | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **8** | 4/10/2025 | Kunal , Dheeraj , Rugved | Final Interface | | Minimal Design | Rugved | ☑ | | |
| **9** | 9/10/2025 | Kunal , Dheeraj,Vilas Sir | Final | | | | ☑ | | |
| **10** | 29/10/2025 | Dheeraj, Kunal | | | | | ☑ | | |
| **11** | 31/10/2025 | Kunal , Dheeraj,Vilas Sir | | | | | ☑ | | |
| **12** | | | | | | | ☐ | | |